

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0492

Cub-Run Dp 138 kV
October 20, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0492. The Interconnection Customer proposes a 447.0 MW solar photovoltaic facility interconnecting at the Cub-Run Dp 138 kV substation in the Virginia area. The request was received on June 25, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 255 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,220,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Del Amo – Grand Blanc Generating Sta 138 kV, the Grand Blanc Generating Station – Walney 138 kV line loads to 179.0% of its 210 MVA rating. Project DFAX: 45.3%.

Cub-Run Dp – Grand Blanc Generating Sta 138 kV: 159.4% loading (346 MVA) following the outage of the Colonial – Idaho Springs 138 kV; DFAX = 52.9%.

For the loss of Del Amo – Grand Blanc Generating Station 138 kV, the Sully – Walney 138 kV loads to 150.6% of its 226 MVA rating. Project DFAX: 45.3%.

For the loss of Cub-Run Dp – Grand Blanc Generating Station 138 kV Line, the Pleasant Vly – Cub-Run Dp 138 kV L loads to 150.4% of its 278 MVA rating. Project DFAX: 100.0%.

For the loss of Cub-Run Dp – Grand Blanc Generating Station 138 kV, the Fountain – Pleasant Vly 138 kV loads to 150.0% of its 266 MVA rating. Project DFAX: 100.0%.

The Del Amo – Bull Run 138 kV is overloaded to 133.7% of its 226 MVA emergency rating for the loss of the Grand Blanc Generating Sta – Walney 138 kV. The project contributes 32.5% DFAX (145.2 MW) to this overload.

Fillmore – Bull Run 138 kV Line: 128.2% loading (219 MVA) following the outage of the Grand Blanc Generating Station – Walney 138 kV; DFAX = 32.5%.

Thomaston – Fountain 138 kV: 127.8% loading (316 MVA) following the outage of the Cub-Run Dp – Grand Blanc Generating Station 138 kV; DFAX = 74.1%.

For the loss of Grand Blanc Generating Station – Del Amo 138 kV, the Castleton Energy Ctr – Sully 138 kV loads to 127.6% of its 249 MVA rating. Project DFAX: 45.3%.

For the loss of Grand Blanc Generating Sta – Cub-Run Dp 138 kV, the Thomaston – Toa Vaca 138 kV L loads to 126.0% of its 159 MVA rating. Project DFAX: 44.3%.

Yavapai – Luna 138 kV L: 124.8% loading (246 MVA) following the outage of the Grand Blnc Generating Sta – Cub-Run Dp 138 kV; DFAX = 10.1%.

For the loss of Grand Blanc Generating Station Substation – Cub-Run Dp 138 kV, the Peavy Fall Hy – Luna 138 kV loads to 119.3% of its 228 MVA rating. Project DFAX: 10.1%.

Del Amo – Grand Blanc Generating Sta 138 kV L: 115.1% loading (300 MVA) following the outage of the Grand Blanc Generating Station – Walney 138 kV; DFAX = 32.5%.

Beltsville – Bunker Hill 138 kV L: 112.4% loading (1,020 MVA) following the outage of the Taber – Glenannan 345 kV; DFAX = 5.4%.

For the loss of Taber – Glenannan 345 kV, the Bunker Hill – Nasa 138 kV loads to 111.4% of its 1,063 MVA rating. Project DFAX: 5.4%.

Siuc – Fairland 138 kV: 109.0% loading (832 MVA) following the outage of the Taber – Glenannan 345 kV; DFAX = 5.5%.

South Houston – Peavy Fall Hy 138 kV Line: 108.6% loading (209 MVA) following the outage of the Grand Blanc Generating Sta – Cub-Run Dp 138 kV; DFAX = 10.1%.

Tokoma – Yulee 138 kV line: 107.4% loading (331 MVA) following the outage of the Darien – Colesville 138 kV line; DFAX = 5.4%.

The Westport – Wheelabrator Baltimore Refuse Substation 138 kV is overloaded to 106.3% of its 882 MVA emergency rating for the loss of the Taber – Dixleta 345 kV line. The project contributes 5.9% DFAX (26.5 MW) to this overload.

Tiger Bay – Dranesville 345 kV: 105.3% loading (1,183 MVA) following the outage of the Glenannan – Taber 345 kV; DFAX = 16.5%.

For the loss of Taber – Glenannan 345 kV, the Riva Rd – Sierra Pacific Lumber 345 kV loads to 105.1% of its 1,495 MVA rating. Project DFAX: 6.8%.

The Sierra Pacific Lumber 345 kV/1 kV Autotransformer is overloaded to 105.1% of its 1,495 MVA emergency rating for the loss of the Taber Substation – Glenannan 345 kV line. The project contributes 6.8% DFAX (30.3 MW) to this overload.

For the loss of Taber – Glenannan 345 kV L, the Sierra Pacific Lumber Substation 138/1 kV TX loads to 105.1% of its 1,495 MVA rating. Project DFAX: 6.8%.

For the loss of Taber – Dixleta 345 kV, the Montpelier – Drum 138 kV loads to 103.8% of its 745 MVA rating. Project DFAX: 5.4%.

Darien – Colesville 138 kV: 102.5% loading (718 MVA) following the outage of the Taber – Glenannan 345 kV; DFAX = 5.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Walney – Grand Blanc Generating Sta Substation 138 kV	Rebuild line with higher-capacity conductor	\$2,758,000	100% (\$2,758,000)
Cub-Run Dp – Grand Blanc Generating Sta 138 kV	Rebuild line with higher-capacity conductor	\$2,807,000	100% (\$2,807,000)
Sully – Walney 138 kV	Rebuild line with higher-capacity conductor	\$2,570,000	100% (\$2,570,000)
Pleasant Vly – Cub-Run Dp 138 kV L	Rebuild line with higher-capacity conductor	\$2,359,000	100% (\$2,359,000)
Fountain – Pleasant Vly 138 kV	Rebuild line with higher-capacity conductor	\$6,503,000	100% (\$6,503,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Del Amo – Bull Run 138 kV	Rebuild line with higher-capacity conductor	\$2,806,000	100% (\$2,806,000)
Fillmore – Bull Run 138 kV	Rebuild line with higher-capacity conductor	\$6,399,000	100% (\$6,399,000)
Thomaston – Fountain 138 kV	Rebuild line with higher-capacity conductor	\$1,274,000	100% (\$1,274,000)
Castleton Energy Ctr – Sully 138 kV	Rebuild line with higher-capacity conductor	\$2,501,000	100% (\$2,501,000)
Thomaston – Toa Vaca 138 kV	Rebuild line with higher-capacity conductor	\$1,207,000	100% (\$1,207,000)
Yavapai – Luna 138 kV	Rebuild line with higher-capacity conductor	\$1,587,000	100% (\$1,587,000)
Peavy Fall Hy – Luna 138 kV L	Rebuild line with higher-capacity conductor	\$1,699,000	100% (\$1,699,000)
Del Amo – Grand Blanc Generating Sta 138 kV L	Rebuild line with higher-capacity conductor	\$1,523,000	100% (\$1,523,000)
Beltsville – Bunker Hill 138 kV	Rebuild line with higher-capacity conductor	\$556,000	3% (\$17,000)
Bunker Hill – Nasa 138 kV	Rebuild line with higher-capacity conductor	\$1,696,000	3% (\$51,000)
Siuc – Fairland 138 kV	Reconductor line	\$628,000	8% (\$49,000)
S Houston – Peavy Fall Hy 138 kV Line	Reconductor line	\$524,000	100% (\$524,000)
Tokma – Yulee 138 kV	Reconductor line	\$531,000	5% (\$24,000)
Wheelabrator Baltimore Refuse – Westport 138 kV	Reconductor line	\$209,000	3% (\$7,000)
Dranesville – Tiger Bay 345 kV L	Reconductor line	\$10,101,000	5% (\$490,000)
Riva Rd – Sierra Pacific Lumber 345 kV	Reconductor line	\$11,078,000	3% (\$354,000)
Sierra Pacific Lumber 345/1kV Autotransformer	Replace transformer with higher-rated unit	\$12,817,000	3% (\$410,000)
Sierra Pacific Lumber Substation 138/1 kV TX	Replace transformer with higher-rated unit	\$4,776,000	3% (\$153,000)
Drum – Montpelier 138 kV L	Reconductor line	\$2,498,000	4% (\$92,000)
Darien – Colesville 138 kV	Reconductor line	\$414,000	14% (\$56,000)
	Total allocated to project		\$38,220,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.